Lesson notes · for you

Questions in SQL

Data engineering · Level 3 · Lesson 2

Time
15 min

Materials
Your phone with Darsi, for the SQL and Python steps; Paper and a pen

Learning goals

- Query with SELECT, WHERE, ORDER BY and LIMIT.
- Summarise with COUNT, AVG and GROUP BY; explain NULL.
- Do the same count in Python with a dictionary.

Activity: Human GROUP BY

1. Cards (3 min): Each pupil gets a card for one made-up loan: a class, a book and a date.
2. WHERE (3 min): Only the 5B cards stand up; then they line up newest first: ORDER BY borrowed DESC.
3. GROUP BY (5 min): Everyone groups by class and counts themselves; write one row per class on the board.
4. Check (4 min): Do the group counts add up to the number of cards? Write the SQL for what you just did.

Prepare the loan cards with made-up data before the lesson.

Discussion

- Why do dates written year-month-day sort correctly?
- When do COUNT(*) and COUNT(mark) give different answers?
- How is a Python dictionary like GROUP BY?

Check yourself

- Write a query for the number of loans of each book, the most first.

Try it this week
Write three questions you'd ask your school library's database. Then write each one in SQL: SELECT, FROM, WHERE, and maybe GROUP BY.

Worksheet

Questions in SQL

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Name _____ Class _____ Date _____

1 **Which line keeps only the rows for class 5B?**

◯ WHERE class = '5B' ◯ GROUP BY class ◯ ORDER BY class

2 **Show the pupil and the book for every loan by class 5B, the most recent first.**

```
SELECT pupil, book FROM loans
WHERE class = '5B';
```

Write your program:

3 **WHERE class = 5B, without quotes, finds the rows for class 5B.**

◯ True ◯ False

4 **The librarian's question: which class borrowed the most? Show each class with its number of loans, the most first.**

```
SELECT class, COUNT(*) FROM loans;
```

Write your program:

5 **Your query says one class borrowed 900 books this term. Which of these are good ways to judge the answer? Choose all that apply.**

☐ Check that the class counts add up to the total number of loans

☐ Run it on a few rows you can count by hand ☐ Trust it, because it ran without an error

☐ Ask whether 900 is even possible for one class

Write one question you'd ask a database at school (the library, the canteen, the sports team). Which SQL parts would it need?